Mechanical black box 2.0 — Solution

Y26 (2026) — English translation

01 **-20.00 pts**

Do not try to open the MCH! For opening the ball, you will be disqualified and the result of the round will be cancelled!

02 **-1.00 pts**

Carefully check the reliability of your installation, especially its moving parts! Repairing the MCH can last up to 30 minutes! Starting from the second, the breakdown of the MCH leads to a penalty of 1 point!

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A0 **0.10 pts**

Write down the number of the mechanical black box issued to you.

A1 **0.10 pts**

Measure and record the values of the mass $(M + m)$ and length L of the MCH.

Answer: \textbfAnswer:

$$(M + m) = 199,2 \text{ g}$$

$$L = 45,0 \text{ cm}$$

A2 **0.90 pts**

Using drawings, describe the measurement method that allows you to obtain the product $m \cdot x_0$ of the mass of the bob and its equilibrium coordinate.

We will find the coordinate of the center of mass by placing the MCN on the edge of the table so that the MCN almost turns over. Let's take 2 measurements. In the first case, we smoothly transfer the MCH from a vertical position with a weight from below to a horizontal one, in the second, from a vertical position with a weight on top. This is necessary to average the static friction force.

Note. An alternative way to eliminate friction is to tap or roll the box on the table to bring the weight into equilibrium.

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$$m \cdot x_0 = (M + m)x_{cm}$$

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